WORKED SOLUTION

STEP I 2008, Question 11

Mechanics • Limiting Friction Equilibrium

STEP I 2008, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q11

The key idea

Because $\angle P_1OP_2$ is a right angle, the radius OP_2 is parallel to the tangent at P_1, and the radius OP_1 is parallel to the tangent at P_2. So if we resolve along the two radii, each equation involves just one normal reaction and one friction force. Add a moment equation about O, where both normal reactions have no moment, and limiting friction $F = \mu R$ at each end gives three equations from which α drops out. The final part is the double-angle formulae in disguise.

Solution

The model and the forces

Treat the rod as a uniform rigid rod with centre of mass G at its midpoint, the rings as light, and let the wire have radius a. Since the rings are light, the wire's force on each ring is passed straight to the rod, so we can treat the rod as touching the wire at P_1 and P_2. At each end there is a normal reaction along the radius and a friction force along the tangent.

Take axes with origin O, x to the right and y upwards, with P_1 to the left of the lowest point of the wire. Since $\angle P_1OP_2 = 90^\circ$ and P_1 makes angle α with the downward vertical, OP_2 makes angle $90^\circ - \alpha$ with it, on the other side:

$$P_1 = a(-\sin\alpha, -\cos\alpha), \qquad P_2 = a(\cos\alpha, -\sin\alpha).$$

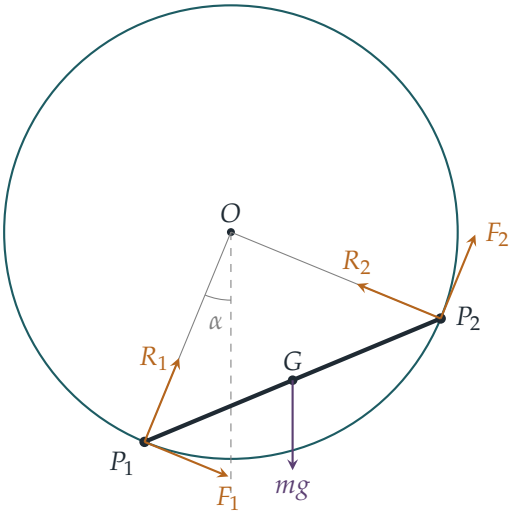
As $0 < \alpha < 45^\circ$, $\cos\alpha > \sin\alpha$, so P_1 is indeed lower than P_2, and both lie below O.

The midpoint is $G = \frac{a}{2}(\cos\alpha - \sin\alpha, -\cos\alpha - \sin\alpha)$, which lies to the *right* of the vertical through O. So the weight has a clockwise moment about O, and the rod is about to slip clockwise round the wire (P_2 moving down, P_1 moving up). Friction opposes this, so it acts *anticlockwise* at both ends: down the wire at P_1 and up the wire at P_2.

Let $\mathbf{u}_1$ and $\mathbf{u}_2$ be the unit vectors from P_1 and P_2 towards O:

$$\mathbf{u}_1 = (\sin\alpha, \cos\alpha), \qquad \mathbf{u}_2 = (-\cos\alpha, \sin\alpha), \qquad \mathbf{u}_1 \cdot \mathbf{u}_2 = 0.$$

The anticlockwise tangent at P_1 is $-\mathbf{u}_2$ and at P_2 it is $\mathbf{u}_1$. Let the normal reactions (towards O) be R_1, R_2 and the friction forces be F_1, F_2.



The equations of equilibrium

The weight mg acts vertically downwards, so its components along $\mathbf{u}_1$ and $\mathbf{u}_2$ are $-mg\cos\alpha$ and $-mg\sin\alpha$.

Resolving along $\mathbf{u}_1$ (that is, along P_1O): R_1 and F_2 contribute, while R_2 and F_1 are perpendicular to it. So

$$R_1 + F_2 = mg\cos\alpha. \tag{1}$$

Resolving along $\mathbf{u}_2$ (along P_2O): R_2 contributes $+R_2$ and F_1, which points along $-\mathbf{u}_2$, contributes $-F_1$. So

$$R_2 - F_1 = mg\sin\alpha. \tag{2}$$

Taking moments about O: the normal reactions pass through O. Each friction force is tangential at distance a from O, giving anticlockwise moments aF_1 and aF_2. The weight acts at horizontal distance $\frac{a}{2}(\cos\alpha - \sin\alpha)$ from O, clockwise. Hence

$$F_1 + F_2 = \tfrac{1}{2}mg(\cos\alpha - \sin\alpha). \tag{3}$$

Limiting friction and $\tan\alpha$

The rod is on the point of slipping at both ends, so $F_1 = \mu R_1$ and $F_2 = \mu R_2$. Writing $c = \cos\alpha$, $s = \sin\alpha$, equations (1) and (2) become

$$R_1 + \mu R_2 = mgc, \qquad -\mu R_1 + R_2 = mgs,$$

with solution

$$R_1 = \frac{mg(c - \mu s)}{1 + \mu^2}, \qquad R_2 = \frac{mg(s + \mu c)}{1 + \mu^2}.$$

Both are positive when μ is small (as it must be here; see below), so the reactions do act towards O as drawn. Now (3) says $\mu(R_1 + R_2) = \frac{1}{2}mg(c - s)$, that is,

$$\begin{aligned} 2\mu(c - \mu s + s + \mu c) &= (1 + \mu^2)(c - s) \\ c\,(2\mu + 2\mu^2 - 1 - \mu^2) + s\,(2\mu - 2\mu^2 + 1 + \mu^2) &= 0 \\ s\,(1 + 2\mu - \mu^2) &= c\,(1 - 2\mu - \mu^2). \end{aligned}$$

Dividing by $c\,(1 + 2\mu - \mu^2)$ (nonzero, since $c > 0$ and $\tan\alpha > 0$ forces μ to be small, as we see next),

$$\tan\alpha = \frac{1 - 2\mu - \mu^2}{1 + 2\mu - \mu^2},$$

as required.

Ask yourself How do we know the friction acts anticlockwise rather than clockwise?

The weight's moment about O is clockwise because G is to the right of O, and the normal reactions have no moment about O. So friction alone must supply an anticlockwise moment, and the rod would slip clockwise without it. If you had drawn friction the other way, you would get the same working with μ replaced by $-\mu$, giving $\tan\alpha > 1$, which contradicts $\alpha < 45°$.

The angle of the rod

The vector from P_1 to P_2 is $a(\cos\alpha + \sin\alpha,\ \cos\alpha - \sin\alpha)$, whose components are both positive. So the acute angle θ that the rod makes with the horizontal satisfies

$$\tan\theta = \frac{\cos\alpha - \sin\alpha}{\cos\alpha + \sin\alpha} = \frac{1 - \tan\alpha}{1 + \tan\alpha} = \tan(45° - \alpha),$$

and since $0 < 45° - \alpha < 45°$, we get $\theta = 45° - \alpha$.

Now put $\mu = \tan\lambda$ in the formula for $\tan\alpha$ and multiply top and bottom by $\cos^2\lambda$:

$$\tan\alpha = \frac{(\cos^2\lambda - \sin^2\lambda) - 2\sin\lambda\cos\lambda}{(\cos^2\lambda - \sin^2\lambda) + 2\sin\lambda\cos\lambda} = \frac{\cos 2\lambda - \sin 2\lambda}{\cos 2\lambda + \sin 2\lambda} = \frac{1 - \tan 2\lambda}{1 + \tan 2\lambda} = \tan(45° - 2\lambda).$$

Since $0 < \lambda < 22.5°$, the angle $45° - 2\lambda$ lies strictly between $0°$ and $45°$, as does α. The tangent function is one-to-one on this interval, so $\alpha = 45° - 2\lambda$. Therefore

$$\theta = 45° - \alpha = 2\lambda,$$

which is what we had to show. (This also explains the condition $\lambda < 22.5°$, i.e. $\mu < \sqrt{2} - 1$: it is exactly what makes $\tan\alpha > 0$.)

Checks

- With $\mu = 0$ (smooth wire) we get $\tan\alpha = 1$, so $\alpha = 45°$ and the rod is horizontal, which is the only position in which a rod on a smooth circle can rest.

- With $\mu = 0.2$: $\tan\alpha = 0.56/1.36$, so $\alpha \approx 22.38^\circ$ and $\theta = 45^\circ - \alpha \approx 22.62^\circ$, which matches $2\arctan 0.2 \approx 22.62^\circ$.
- Numerically, for 10^4 random values of $\mu \in (0, \sqrt{2}-1)$, a and m, substituting these R_1, R_2, F_1, F_2 into the full vector equations gave zero resultant force and zero moment (to about 10^{-14}), with $R_1, R_2 > 0$ and $\theta = 2\lambda$.

Common slips

- Resolving horizontally and vertically. It works, but every equation then contains all four forces; resolving along the two perpendicular radii is much cleaner.
- Getting the direction of friction wrong at one end. Both rings slip the same way round the circle, so friction is anticlockwise at *both* ends.
- Forgetting that $\theta = 45^\circ - \alpha$ needs a word of justification: the chord is perpendicular to the bisector of $\angle P_1 O P_2$.

Beyond STEP

There is a slicker geometric view: at limiting friction the total reaction at each end makes angle λ with the radius, and a body in equilibrium under three forces needs them to be concurrent, which leads straight to $\theta = 2\lambda$. This "angle of friction" idea, and the fact that friction is an inequality $|F| \leq \mu R$ rather than an equation, are the starting points for statics problems with several contacts, where you must ask which contacts slip first. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body equilibrium with this kind of care.

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